

EDUCATIONAL BACKGROUND

- Sep/2025 - Current. **University of Chicago** Chicago, IL, USA
M.A. in Computational Social Science GPA: 3.83/4.00
- Jan/2021 - Jun/2025. **University of California, San Diego** San Diego, CA, USA
B.S. in Cognitive Science (Machine Learning Track) with Highest Distinction GPA: 3.71/4.00
- *B.S. in Applied Mathematics*
Minor in History; Minor in Linguistics

Related Courses: Neural Networks and Deep Learning (A+); Computational Linguistics (A+); Machine Learning with Few Labels (A); Advanced Machine Learning Methods (A+); Modeling & Data Analysis (A).
Honors: **Provost Honors** (6 terms), **Triton Research & Experiential Learning Scholars** (2 terms)

RESEARCH INTERESTS

I study language models from an NLP and cognitive science perspective: how their behavior aligns—or fails to align—with what people expect from a competent communicator, especially in ambiguous, creative, or knowledge-laden contexts. I am drawn to questions where standardized benchmarks miss what people actually care about, and to building human-grounded evaluations that surface these gaps.

PUBLICATIONS & PREPRINTS

- **Zehan Li**, Yutong Zhu, Siyang Wu, Honglin Bao, James A. Evans. Narrative Flattening: How Post-Training Compresses Thematic, Affective, and Stylistic Variation in LLM Fiction. *Accepted to EMNLP 2026*. arXiv preprint: 2605.27878.
- **Zehan Li**, Yuxuan Wang, Ali El Lahib, Ying-Jieh Xia, Xinyu Pi. Simulated Ignorance Fails: A Systematic Study of LLM Behaviors on Forecasting Problems Before Model Knowledge Cutoff. *Accepted to IJCAI 2026, Main Conference*. arXiv preprint: 2601.13717.
- Ali El Lahib, Ying-Jieh Xia, **Zehan Li**, Yuxuan Wang, Xinyu Pi. Temporal Leakage in Search-Engine Date-Filtered Web Retrieval: A Case Study from Retrospective Forecasting *Accepted to ACL 2026, Main Conference*. arXiv preprint: 2602.00758.
- **Zehan Li**, Ruhua Pan, Xinyu Pi. Beyond LLMs: A Linguistic Approach to Causal Graph Generation from Narrative Texts. *Proceedings of the 7th Workshop on Narrative Understanding (WNU @ NAACL 2025)*. ACL Anthology.

RESEARCH EXPERIENCES

- Why do LLM-generated stories feel narratively flat? Jan/2026 - Current
Project Lead. Supervised by Dr. James A. Evans, Knowledge Lab, University of Chicago
 - Led a **matched story-continuation study examining how different post-training stages reshape LLM-generated fiction**, comparing model against human continuations across multiple story corpora.
 - Developed computational measures of thematic motion, affective prevalence, and linguistic diversity, showing that post-training progressively compresses narrative variation into more uniform, neutral narrower outputs.
- Do language models exacerbate inequalities in scientific attention? Oct/2025 - Current
Researcher. Supervised by Prof. Bernard Koch, University of Chicago
 - Developed a **hierarchical generative model decomposing LLM citation amplification** into training data selection, retrieval bias, and substitution bias, enabling precise attribution of where inequality originates.
 - Designed a citation in-painting task as a realistic use case, running retrieval experiments across Gemini, Claude, GPT, and DeepSeek on ~100K papers sampled from SciSciNet v2 with stratified inverse-probability weighting.
 - Used Infini-gram to obtain exact training-corpus frequencies, enabling within-frequency-stratum comparisons that isolate model behavior from data composition effects.

- Can we automate the generation of mechanistic interpretability reports? Sep/2025 - Current
Research assistant. Supervised by Prof. James A. Evans, University of Chicago
 - **Built an automated framework for generating structured mechanistic interpretability reports**, integrating a lightweight RAG stack to retrieve papers, construct citation graphs, and produce literature packs for contextualized analysis.
 - Engineered an end-to-end execution system that exports publishable Markdown reports with original analyses and figures, enabling scalable and reproducible interpretability research.
- Can prompts reliably rewind LLM knowledge for forecasting evaluation? Jul/2025 - Jan/2026
Project Lead. Supervised by Prof. Zhiting Hu, UCSD
 - Resulting publications: 2601.13717 and 2602.00758 . **Provided the first systematic test of whether Simulated Ignorance (SI) can approximate True Ignorance (TI) in retrospective LLM forecasting evaluation**, exposing fundamental flaws in existing forecasting benchmarks.
 - Independently conceptualized the experimental framework and deployed Qwen3-32B and Llama models on cluster servers. Across 477 competition-level questions and 9 models, found that SI fails systematically: cutoff instructions leave a 52% performance gap, and chain-of-thought reasoning cannot suppress implicit post-cutoff knowledge.
 - Discovered that reasoning-optimized models exhibit *worse* SI fidelity despite superior reasoning trace quality, demonstrating that prompts cannot reliably “rewind” model knowledge states. Recommended against using SI-based retrospective setups for benchmarking forecasting capabilities.
- Quantifying Conceptual Density and vacuity in Text Aug/2024 - Jun/2025
Undergrad Honors Thesis. Supervised by Prof. Zhuowen Tu & Alex Warstadt, UCSD
 - **Developed a theoretical model for quantifying text specificity and vacuity** using linguistic properties as primary dimensions, proposing a metric system for evaluating conceptual density in written communication.
 - Benchmarked outputs from multiple LLMs to compare their conceptual density and informational richness against human-written text.
 - Incorporated human cognitive metrics by estimating daily concept and category exposure to connect text specificity with cognitive engagement.
- Can linguistic properties improve causal graph generation from narratives? Feb/2024 - Sep/2024
Project Lead. Supervised by Prof. Zhiting Hu, UCSD
 - Resulting publication: WNU @ NAACL 2025. **Developed an open-source linguistics-based model for causal graph generation**, achieving 82% interpretability, a 9% improvement over direct GPT prompting baselines.
 - Built a feature extraction pipeline using spaCy and nltk that leverages linguistic properties (e.g., time boundness) for causal link identification, yielding a 15% accuracy gain over LLM-only approaches based on human evaluation.
 - Independently authored a peer-reviewed paper detailing methods for efficient and interpretable causal inference in narrative texts.

EXTRACURRICULAR EXPERIENCES

- Metaculus AI Forecasting Tournament (Fall 2025 Cycle) Oct/2025 - Dec/2025
 - Developed an autonomous AI forecasting agent that competed in the Metaculus AI Bot Tournament, achieving **15th place globally** and a \$1,500 prize.
 - Engineered a well-calibrated probabilistic prediction system, demonstrated by close alignment between predicted confidence and actual resolution rates across the full probability range.

PROFESSIONAL EXPERIENCES

- Instructional Assistant (2 courses): Deep Learning & Supervised Machine Learning Sep/2024 - Jun/2025
UCSD Cognitive Science Department. Under Prof. Zhuowen Tu
- Led weekly 80-minute discussion sessions for 100+ students across two quarters, covering topics from regression and classification to GPT architectures and prompt engineering.
 - Provided individualized support through office hours and detailed assignment feedback.

OTHER INFORMATION

- Computer Skills: Python, Java, SQL, R, MATLAB, JavaScript, L^AT_EX, HTML/CSS, Docker
- Languages: Chinese (native), English (proficient), French(Intermediate)